



MATHEMATICS UNIT
KOLEJ MATRIKULASI KEJURUTERAAN PAHANG

MATHEMATICS 1

Name : _____

Practicum : _____

Lecture's Name : _____

INSTRUCTIONS TO STUDENTS:

1. This paper consists of 2 sections. Section A consists of 3 questions and Section B consists of 7 questions.
2. Answer **all** questions.
3. The full mark for each question is shown in the bracket at the end of each of the question.
4. All workings must be shown clearly.
5. Only non-programmed scientific calculators can be used.
6. Numerical answers can be given in the form of π , e , surd, fractions or up to three significant figures, where appropriate, unless stated otherwise in the question.

QUESTION	SECTION A	SECTION B
1	/ 9	/ 5
2	/ 9	/ 11
3	/ 7	/ 6
4		/ 14
5		/ 14
6		/ 16
7		/ 9
TOTAL FOR SECTION	/25	/75
TOTAL / 100		

LIST OF MATHEMATICAL FORMULAE**Trigonometry**

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \sin^2 A - \cos^2 A$$

$$= 2 \cos^2 A - 1$$

$$= 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

LIST OF MATHEMATICAL FORMULAE**Differentiation**

$f(x)$	$f'(x)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

The Sum Rule $\frac{d}{dx}(u(x) + v(x)) = \frac{du}{dx} + \frac{dv}{dx}$

Product Rule $\frac{d}{dx}(u(x) \cdot v(x)) = u \frac{dv}{dx} + v \frac{du}{dx}$

Quotient Rule $\frac{d}{dx}\left(\frac{u(x)}{v(x)}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

Chain Rule $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$

$$\frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right) \cdot \frac{dt}{dx}$$

Shape	Volume	Surface Area
Sphere	$V = \frac{4}{3}\pi r^3$	$S = 4\pi r^2$
Right circular cone	$V = \frac{1}{3}\pi r^2 h$	$S = \pi r^2 + \pi r h$
Right circular cylinder	$V = \pi r^2 h$	$S = 2\pi r^2 + 2\pi r h$

SECTION A [25 marks]

This section consists of 3 questions. Answer **all** questions.

1 a) If $\lim_{x \rightarrow 1} \frac{5x^2 - px}{2x^2 - x} = 1$, find the value of p. [3 marks]

b) Evaluate $\lim_{x \rightarrow +\infty} \left(\frac{8x^2 + 8x - 2}{x^2 + 1} \right)^{\frac{1}{3}}$. [3 marks]

c) Find $\lim_{x \rightarrow 2^-} \frac{(3x + 2)|2 - x|}{2 - x}$. [3 marks]

2 Find $\frac{dy}{dx}$ for each of the following:

a) $y = 3x^2 \ln(2x^3 - 5)^4$ [4 marks]

b) $y = \frac{1 + \cos x}{\sqrt{1 + \sin 3x}}$. [5 marks]

3 A ship is 6 km east and 8 km north of a lighthouse. It is moving north at a rate of 13 km/h and east at a rate of 16 km/h. The movement of the ship from lighthouse can be represented as a right-angle triangle. At what rate is the ship's distance from the lighthouse changing?

[7 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answer **all** questions.

- 1 Given that the complex number $z = \sqrt{-1}(2i - i^{10})$, express z in Cartesian form.
Hence, find the modulus of z . [5 marks]

- 2 Solve
 - a) $1 + \sqrt{x+3} = \sqrt{2x-1}$ [5 marks]
 - b) $\left| \frac{x+2}{x-4} \right| < 5$. [6 marks]

- 3 It is given that $f(x) = 2 - \ln|3-x|$.
 - a) State the domain and range of $f(x)$. [2 marks]
 - b) Find $f^{-1}(x)$ if it exists. [4 marks]

- 4 a) Let f and g be two functions. Given that $g \circ f(x) = \frac{x+2}{3}$ and $g(x) = 2x+5$,
find $f(x-1)$. [5 marks]

- b) Given that $h(x) = \begin{cases} 4e^x, & x < 0 \\ \frac{3}{4}x + 4, & 0 \leq x < 4 \\ x^2 - 8x + 18, & x \geq 4 \end{cases}$.
 - i) Sketch $h(x)$. [5 marks]
 - ii) Find the domain and range for $h(x)$ and $h^{-1}(x)$. [4 marks]

- 5 a) Find the vertical and horizontal asymptotes of $f(x) = \frac{x+4}{\sqrt{x^2-25}}$. [9 marks]

- b) Given the function $h(x) = \begin{cases} e^x + 2, & x < 0 \\ x^2 + 3, & x \geq 0 \end{cases}$.
Determine the continuity of $h(x)$ at $x = 0$. [5 marks]

- 6 a) Given $y = \frac{x}{2x+1}$. Find
- i) $\frac{dx}{dy}$ [4 marks]
- ii) $\frac{d^2x}{dy^2}$ in terms of y and x . [4 marks]
- b) Given the parametric equation for the curve $x = \frac{2}{\sqrt{2-t^2}}$ and $y = \frac{\sqrt{2-t^2}}{t}$.
Find $\frac{dy}{dx}$. Evaluate $\frac{dy}{dx}$ when $t = 1$. [8 marks]
- 7 A circle and a square is to be formed by using a wire with total length 8 cm. The radius of the circle is x cm while one of the sides of the square is y cm.
- a) By considering the perimeter of both shapes, find an expression for y in terms of x . [2 marks]
- b) Show that the total area of the two shapes is given by
- $$A = \left(\frac{\pi^2 + 4\pi}{4} \right) x^2 - 2\pi x + 4. \quad [3 \text{ marks}]$$
- c) If the total area of the two shapes is minimum possible, what percentage of the wire is used for the circle? [4 marks]

END OF QUESTION PAPER